

CHAPTER 3

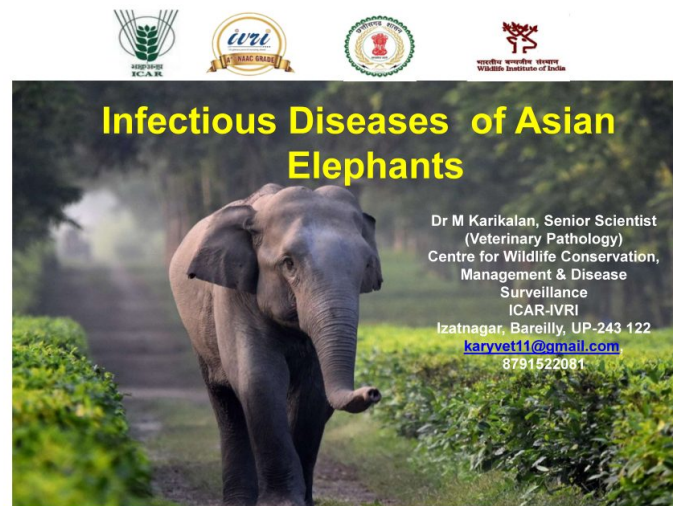
Infectious Diseases of Asian Elephants

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Workshop on Elephant Mortality Investigation and Management

Chhattisgarh Forest Department · Bilaspur · June 2026



The need to conserve !!

- Current population status of Asian elephants shows a declining trend (Williams *et al.*, 2020)
- Asian elephant: categorised as "Endangered" by IUCN
- Assigned under Schedule 1 of wildlife protection act 1972, conferring highest protection
- Listed as Appendix 1 species by CITES
- Population of Asian elephants in India as per the latest elephant census is 22,446 elephants in the wild (MoEF & CC, 2025)



National heritage animal of India

- Habitat loss and fragmentation
- Illegal trade
- Poaching
- Human elephant conflict

(Sakumar *et al.*, 2016; Menon *et al.*, 2017)

NE	DU	LC	NT	VU	< EN >	CR	EW	EX
Kingdom	Animalia							
Phylum	Chordata							
Class	Mammalia							
Order	Proboscidae							
Family	Elephantidae							
Subfamily	Elephantinae							
Genus	Loxodonta							
Species	Loxodonta cyclotis (African forest elephant)							
	Loxodonta africana (African bush elephant)							
Genus	Elephas							
Species	Elephas maximus (Asian elephant)							
Subspecies	Elephas maximus sumatransus							
	Elephas maximus indicus							
	Elephas maximus maximus (Roca <i>et al.</i> , 2015)							

Source presentation: Infectious Diseases of Asian Elephants (Dr. M. Karikalan, 2026)

Overview

- **Species:** Asian Elephant (*Elephas maximus*) — Endangered (IUCN)
- **Wild population:** ~22,446 individuals across 13 range countries
- **India:** Largest population; Schedule I (Wildlife Protection Act 1972); CITES Appendix I
- **Scope of lecture:** Infectious, parasitic and management-related diseases — diagnosis, treatment and prevention in wild and captive populations
- **Lecture series:** VBSC SH_1_3 — compiled and presented for field veterinarians and forest officers

1. Introduction

Wildlife disease investigation has emerged as a critical pillar of conservation medicine. The Asian elephant, already classified as Endangered on the IUCN Red List, faces an escalating burden of infectious and non-infectious diseases, many of which are poorly understood in wild populations. Mortality from preventable or treatable diseases represents a significant threat that can be mitigated through timely diagnosis, skilled post-mortem examination, and preparedness of field veterinarians.

1.1 Epidemiological Terminology

A clear understanding of disease epidemiology is essential for fieldwork. Key terms used throughout this chapter are defined below:

Term	Definition
Emerging Disease	A disease appearing in a population for the first time, or rapidly increasing in incidence/geographic range
Re-emerging Disease	A previously known disease that has reappeared after a period of decline, often in new geographic areas or host species
Transboundary Disease	Infectious disease that spreads across national or ecosystem boundaries, requiring multi-jurisdictional response (e.g., FMD, HS)
Zoonosis	Disease transmissible between vertebrate animals and humans (e.g., TB, Rabies, Anthrax, Leptospirosis)
Spillover	Transmission of a pathogen from a reservoir host to a new (incidental) host species
Pathogenicity	Ability of a pathogen to cause disease in the host
Virulence	Severity of disease caused by a pathogen; degree of harm inflicted on the host

1.2 Taxonomy of the Asian Elephant

Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Proboscidea
Family	Elephantidae
Genus	Elephas
Species	Elephas maximus
Subspecies	E. m. indicus (Indian), E. m. maximus (Sri Lankan), E. m. sumatranus (Sumatran), E. m. borneensis (Borneo)

1.3 Spectrum of Disease

Elephant diseases are broadly categorised into infectious and non-infectious conditions. This distinction is fundamental for triage at a casualty site — while infectious conditions require diagnostic sampling and biosafety precautions, non-infectious causes (trauma, electrocution, nutritional deficiency) require different response protocols.

Infectious Diseases	Non-Infectious Conditions
Viral: EEHV, Rabies, FMD, Poxvirus, EMCV	Electrocution (illegal/accidental)
Bacterial: Tuberculosis, Haemorrhagic Septicaemia, Anthrax, Clostridial infections, Leptospirosis	Drowning / Water body falls
Parasitic: Helminths (Fasciola, Strongyles, Amphistomes), Protozoa (Balantidium, Haemoparasites)	Trauma / Fall from height
Fungal: Aspergillosis (captive)	Obesity and nutritional deficiency
	Inter-elephant conflict / musth-related injury
	Captive management failures (chaining, overwork)

2. Elephant Endotheliotropic Herpesvirus (EEHV)

EEHV is the single most important viral disease in elephants. It causes acute haemorrhagic disease characterised by a peracute clinical course — an apparently healthy calf can die within 24–72 hours of first signs. Understanding EEHV biology, diagnosis, and treatment is therefore a priority competency for any field or zoo veterinarian working with elephants.

2.1 Classification and Subtypes

EEHV belongs to the family *Herpesviridae*, subfamily *Betaherpesvirinae*. In Asian elephants, the clinically important subtypes are:

EEHV Subtypes Relevant to Asian Elephants

- **EEHV-1A / 1B** — Most virulent; responsible for majority of fatal cases in Asia and North America
- **EEHV-4** — Second most common in Asian elephants; documented fatal cases
- **EEHV-5** — Occasionally detected; less frequently fatal
- **EEHV-2, 3, 6, 7** — Primarily documented in African elephants; lower risk in Asian species
- **Carrier state:** Adult elephants are latent carriers with virus persisting in nerve ganglia; disease triggers in calves when maternal antibody wanes (between 1–18 years of age)

Figure 2.1 — Herpesvirus family classification and EEHV phylogenetic tree



Clinical signs observed

- Lethargy/ restlessness
- Rise in body temperature
- Reduced appetite
- Salivation
- Swelling of temporal gland and eye lids
- Cyanosis of tongue
- Abdominal colic
- Recumbency

Death with in 6 to 24 hours

Pic courtesy: Nandankanan Zoo Authorities

2.2 Historical Timeline and India Cases

EEHV haemorrhagic disease (EEHV-HD) was first recognised in 1995 at the Smithsonian National Zoo, Washington DC, in an Asian elephant calf. Since then, multiple fatal cases have been documented globally and in India:

Year	Event / Location
1995	First EEHV-HD case recognised — Smithsonian National Zoo, USA
2007	First confirmed EEHV case in India — Kerala (retrospective diagnosis)
2008–2015	Cases documented in Tamil Nadu (Anamalai Tiger Reserve, Mudumalai)
2021	Chhattisgarh: EEHV-1A positive detected (qPCR from calf death)
2023	Chhattisgarh: Second positive case — active surveillance programme initiated
2024	mRNA-based EEHV vaccine enters trial phase — Houston Zoo, USA

Figure 2.2 — EEHV phylogenetic relationships and fatal haemorrhagic disease history



2.3 Pathomechanism and Vulnerable Age Groups

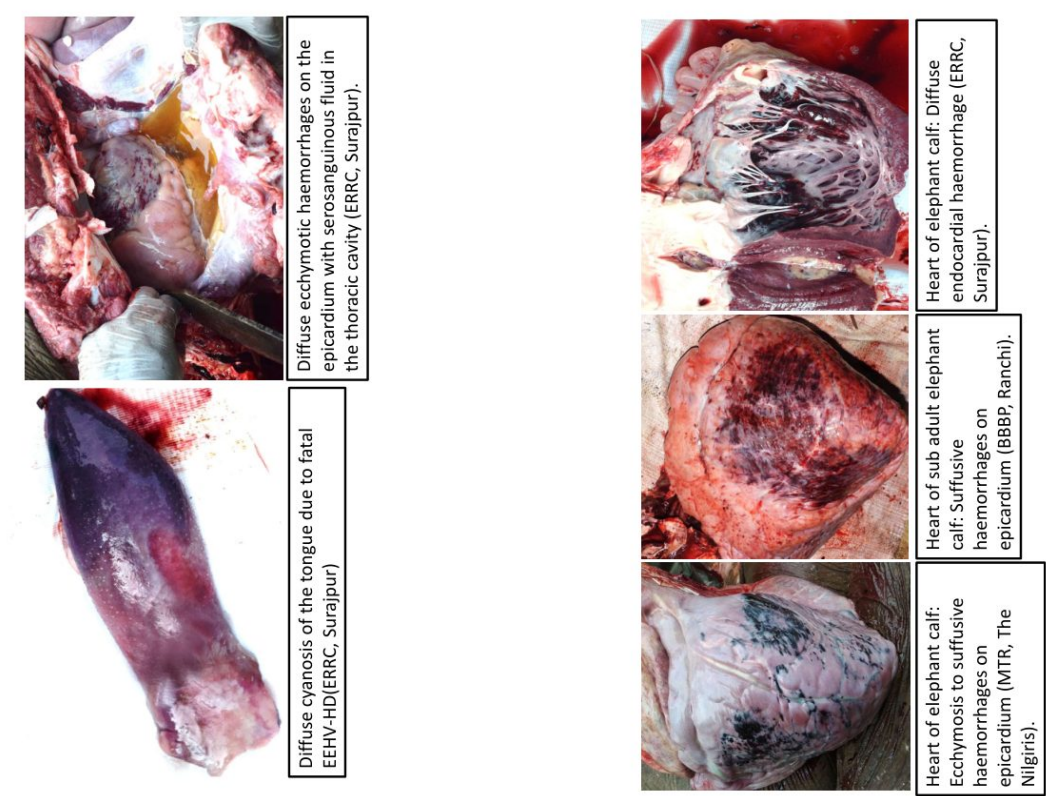
The host preference of EEHV is age-dependent. Calves between 1 and 18 years are the primary victims of fatal disease. Adult elephants harbour the virus as latent carriers but rarely develop clinical disease. The pathomechanism involves endothelial cell tropism — the virus infects cells lining blood vessels, causing massive vascular damage:

Pathomechanism Cascade

- 1. Viral entry:** EEHV infects endothelial cells of capillaries and venules

2. **Vascular damage:** Endothelial lysis → capillary leakage → haemorrhage into tissues
3. **Cytokine storm:** Systemic inflammatory response → hypotension → shock
4. **Organ failure:** Myocardial, hepatic and renal involvement → death within hours to days
5. **Gross lesions:** Epicardial haemorrhages, tongue cyanosis, oedema of face/tongue, petechiation

Figure 2.3 — EHV host preference by age group and pathomechanism cascade



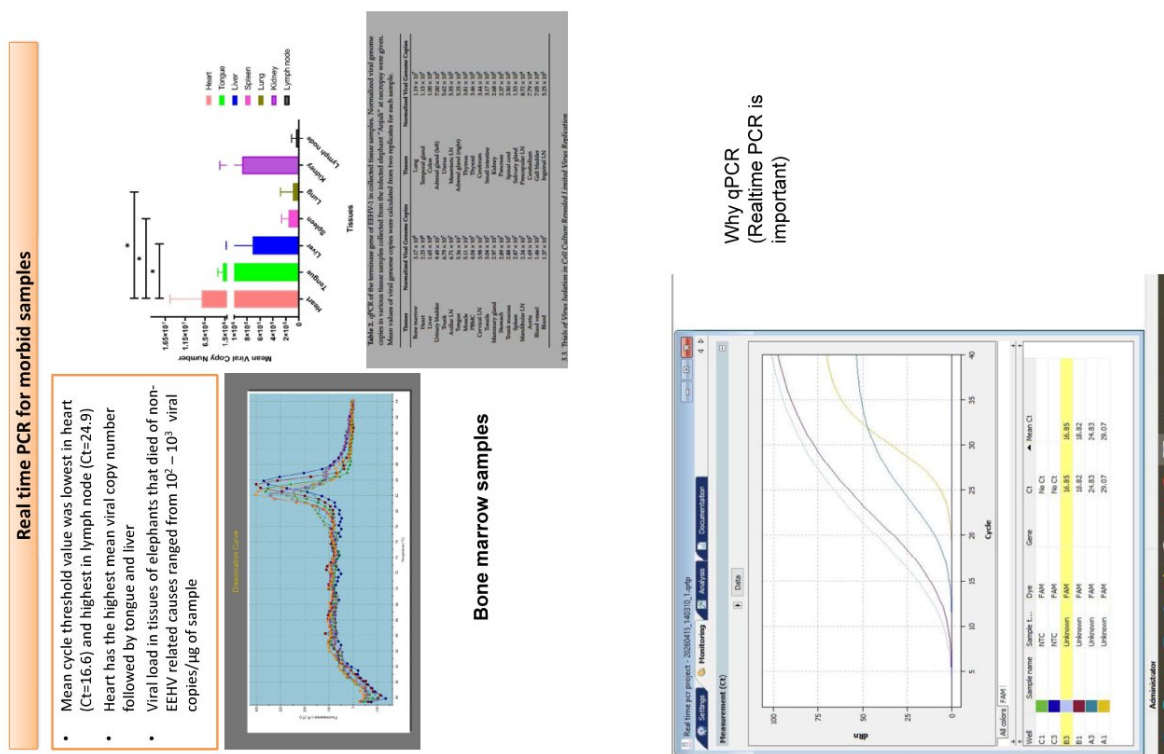
2.4 Clinical Signs

EEHV-HD presents as a peracute haemorrhagic syndrome. The time from first clinical signs to death may be as short as 12–36 hours. The following signs have been documented in confirmed cases:

Clinical Signs of EEHV-HD

- **Generalised oedema** of the head, face, and neck — especially dramatic periorbital and submandibular swelling (hallmark sign)
- **Tongue cyanosis:** Blue-purple discolouration of the tongue mucosa (visible in live animal)
- **Open-mouth breathing** and respiratory distress due to oropharyngeal oedema
- **Weakness, recumbency,** reluctance to walk — rapid deterioration to lateral recumbency
- **Pyrexia** (fever $>38.5^{\circ}\text{C}$) often the first sign detected during routine morning observation
- **Tachycardia** and hypotension in late stages
- **Bloody discharge** from natural orifices may occur terminally
- Death within 24–72 hours of first signs in untreated or late-detected cases

Figure 2.4 — Clinical photographs: EEHV-HD in a calf (Nandankanan Zoological Park)



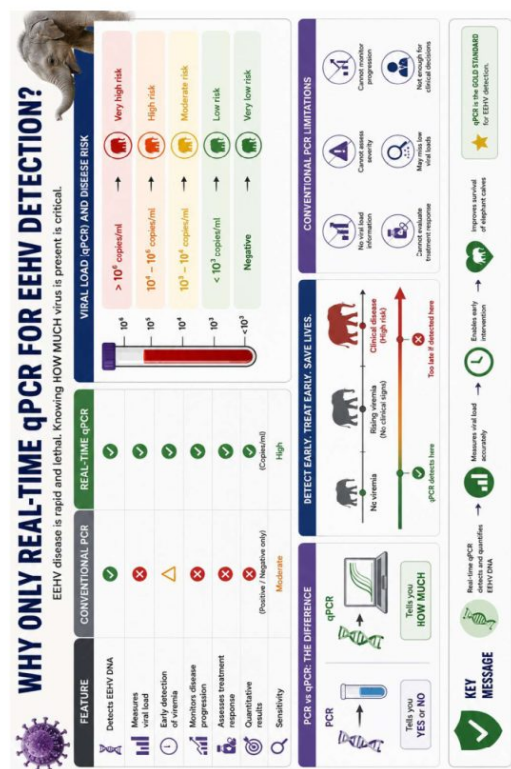
Left: Facial/periorbital oedema in affected calf. Right: Recumbency and open-mouth breathing.

2.5 Post-mortem Findings

Necropsy of EEHV-HD cases reveals characteristic gross and histopathological changes. A thorough and documented post-mortem examination is essential for laboratory confirmation and to rule out other causes of sudden death in young elephants:

Gross Post-Mortem Lesions
• Tongue: Diffuse cyanosis/necrosis; villous thickening; dark blue-black discolouration on cut section
• Heart: Epicardial and endocardial petechiae and ecchymoses; myocardial pallor and mottling
• Liver: Congestion; occasional petechiae; hepatocellular swelling
• Lung: Congestion and oedema; interstitial haemorrhage; frothy exudate in airways
• Lymph nodes: Markedly enlarged, haemorrhagic; mesenteric LN especially affected
• Kidney: Cortical pallor, perivascular haemorrhage
• Spleen: Usually unremarkable or mildly enlarged
• Brain: Usually negative (important for rabies differential exclusion)

Figure 2.5 — PM photographs: tongue cyanosis (ERRC Surajpur), epicardial haemorrhages (MTR Nilgiris), and myocardial lesions (BBBP Ranchi, Nandankanan)



Treatment of EEHV-HD

- Start immediately without waiting for laboratory confirmation
- Human antiviral drugs : used to treat animals with clinical illness (Schmitt et al., 2000; Cracknell, 2008; Wiedner et al., 2012)
- Per-rectal administration of famciclovir @5.5-8.0 mg/kg BID (Schmitt et al., 2001; Tociłowski et al., 2015; Howard et al., 2015; Long et al., 2015; Fuery et al., 2016)
- Initial IV administration followed by oral therapy of ganciclovir @5mg/kg twice daily (Stanton et al., 2013; Padilla et al., 2015)
- Acyclovir @12mg/kg BID for 15 days along and penicillin G (Boonprasert et al., 2015)
- Oral acyclovir therapy (Khammeesri et al., 2020)
- Famciclovir @ 8-15 mg/kg body weight QID on 1 st day and then BID for 3 weeks orally or rectally (Famciclovir is the first drug of choice)
- **Fluid therapy**
 - IV and rectal administration of fluid
 - Per rectal fluid administration: 10-15 litres depending on the clinical status of the animal @ 20ml per kg body weight
 - Plasma therapy
 - Diuretics: Management of peripheral edema - Furosemide 0.6 mg/kg bw, IV or IM QID
 - Antibiotics, anti-inflammatory drugs and immuno-modulators (Levamisole-75 S/C (single dose @ 1ml/50 kg), Vitamin-C)



2.6 Histopathology

Histopathological examination confirms the endotheliotropic nature of EEHV. Representative tissues should be fixed in 10% neutral buffered formalin immediately at necropsy. Characteristic findings on H&E; staining include:

Histopathological Findings (H&E;)
• Tongue mucosa: Intranuclear inclusion bodies in endothelial cells; mucosal ulceration; sub-epithelial haemorrhage; intravascular fibrin thrombi
• Myocardium: Perivascular lymphocytic infiltration; myocyte necrosis; endothelial swelling with inclusion bodies (H&E; ×200)
• Pulmonary sinusoids: Endothelial inclusions; alveolar capillary congestion; occasional fibrin thrombi
• Renal vessels: Glomerular and interstitial haemorrhage; endothelial inclusions
• Diagnostic significance: Intranuclear inclusions are pathognomonic for herpesvirus; their presence in multiple tissues confirms EEHV-HD

Figure 2.6 — Histopathology: H&E; sections (×200) from EEHV-HD cases

Monitoring disease, prevention and control

- Vaccines are still not available (recombinant vectored vaccines, subunit vaccines???)
- New or rescued animal should be properly quarantined before mixing with an existing herd.
- Animal should be quarantined for a period of 6 months at a separate facility with periodical screening of blood sample and trunk wash by PCR besides regular monitoring of vital signs
- Regular close monitoring of animal, visual inspection of different parts of the body, vital signs and behavioral pattern
- CBC, serum biochemical analysis to assess any abnormal haematological changes like leucocytosis, monocytosis, lymphopenia and thrombocytopenia
- Blood sample and trunk wash should be screened at weekly interval against virus through PCR to suspect/confirmed positive elephant
- Screening protocol should be in place for all zoo/private/camp having captive elephant to identify carrier/ shedder in herd
- In pregnant females only periodical monitoring of viral load through q-PCR from the samples like conjunctival swab/ trunk wash/ vaginal wash may be carried out at three-month intervals
- Proper biosecurity measures

ZOO NEWS BLOG

Houston Zoo Elephant Receives First Ever mRNA EEHV Vaccine

June 18, 2024



mRNA vaccines

Recombinant vaccines

Tess, a 40-year-old Asian elephant at Houston Zoo, has been given the first-ever dose of an mRNA vaccine created by virologists at Baylor College of Medicine (BCM) to prevent the deadly elephant endotheliotropic herpesvirus (EEHV) 1A—a devastating viral disease in Asian elephants worldwide. Over the next few weeks, animal care experts and veterinary staff at Houston Zoo will closely monitor Tess to learn how effective this vaccine is for elephants.

Clockwise from top-left: Tongue mucosa, myocardium, pulmonary sinusoids, renal vessels — showing intranuclear inclusion bodies and vascular damage (H&E; ×200)

2.7 Laboratory Diagnosis: qPCR

Quantitative PCR (qPCR) is the gold standard for EEHV diagnosis, both in live animals (antemortem) and at necropsy (postmortem). Understanding the CT value interpretation is essential for field veterinarians:

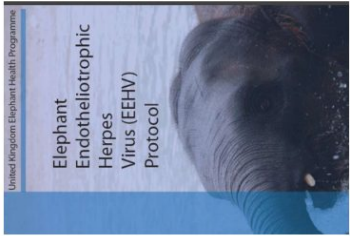
Sample Type	Use	Interpretation
Whole blood (EDTA)	Antemortem surveillance; monitoring in sick calves	CT <30: Active viraemia (treat immediately); CT 30–37: Low-level positive / carrier; CT >37: Negative or below detection
Bone marrow aspirate	Best postmortem sample; highest viral load in active EEHV-HD	Highest sensitivity — preferred for PM diagnosis
Trunk wash	Detection of viral shedding in carriers/adults	Used for herd surveillance; lower sensitivity than blood PCR
Heart / Tongue / Lymph node	PM tissue samples for qPCR and histopathology	Heart and tongue show highest viral copy numbers in EEHV-HD
Liver / Lung / Kidney	Supplementary PM tissues	Variable viral loads; important for severity assessment

Figure 2.7 — Tissue viral load data (qPCR) and qPCR vs conventional PCR comparison

Samples: Blood, serum, trunk wash fluids, oral and rectal swab, tissues, Bone marrow

Standard Operating Procedure for Elephant Endotheliotropic Herpes Virus - Hemorrhagic Disease

CENTRAL ZOO AUTHORITY
Statutory body of the Ministry of Environment, Forest and Climate Change, Government of India
2021



- IVRI, Izatnagar, Bareilly
- Veterinary College, Guwahati
- Centre for Wildlife Health, OUAT, Bhubaneswar,
- Centre for Wildlife Forensic & Health, Jabalpur
- EEHV Diagnostic laboratory, Kerala Forest Department, Wayanad
- State Institute For Animal Diseases, Palode, Kerala

Rabies

- ☐ Rabies is a viral zoonotic neuro invasive disease which causes inflammation in the brain and is usually fatal
- ☐ Transmission mainly through wild canines, Rodents such as mice , rats, squirrels and chipmunks and **small carnivores (Mongoose)**
- ☐ All warm blooded animals are susceptible .
- ☐ The virus can not live outside its host for more than a couple of seconds but has been found in animals as long as 48 hrs after death.
- ☐ In India this disease was mainly reported in captive wild animals of various zoos like **Asian elephants (both in captive and free range)**.
- ☐ Transmitted mainly through the bite from infected animals.
- ☐ Virus may be present in saliva before the onset of clinical signs
- ☐ Variable IPd of 1 week to 1 year
- ☐ Dumb or the furious form
- ☐ Death occurs within 10 days of the 1st symptom

2.8 Treatment Protocol

EEHV-HD is a veterinary emergency. Treatment must begin immediately on clinical suspicion — do not wait for PCR results. Early intervention dramatically improves survival odds. The treatment

protocol used at leading elephant centres in India:

EEHV Treatment Protocol (Field & Zoo)

- **Famciclovir** 5.5–8 mg/kg BID orally — antiviral prodrug (converted to penciclovir); inhibits viral DNA polymerase; first-line agent; must be pre-positioned at elephant camps
- **Acyclovir** IV — used in severe cases when oral famciclovir cannot be administered; acts at same target as famciclovir; give via slow IV infusion
- **Fluid therapy:** Isotonic crystalloids (Ringer's Lactate, Normal Saline) via IV; volume resuscitation for vascular leakage; monitor urine output
- **Anti-inflammatory:** Dexamethasone (cautious use) or Meloxicam for inflammatory cascade
- **Gastric protectants:** Omeprazole / Pantoprazole — prevent stress-related gastric lesions
- **Supportive care:** Shade, quiet environment, soft browse/fruit, mahout presence to reduce stress
- **Monitoring:** Temperature every 4 hours; blood qPCR every 24–48 hours; decreasing CT value = improving; CBC and chemistry every 48 hours if possible

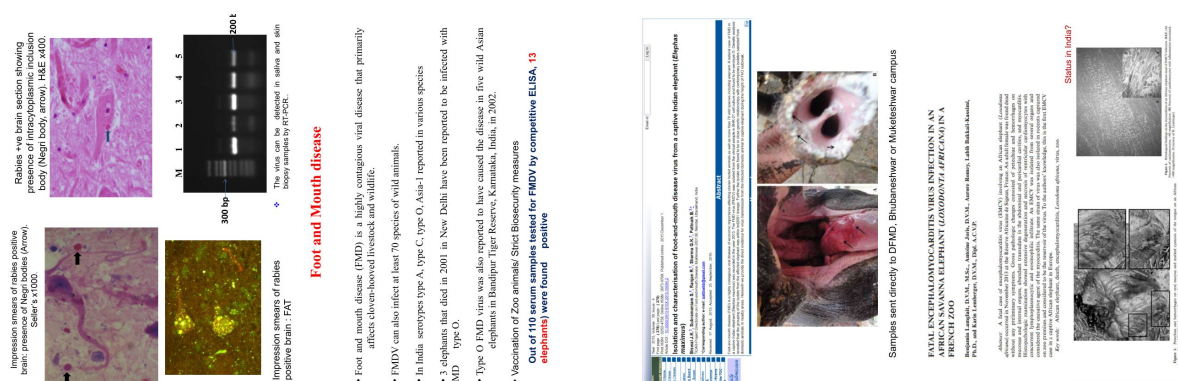
CRITICAL: EEHV Emergency Response

Famciclovir and Acyclovir must be pre-positioned at all elephant camps and zoos housing calves. Delay in starting antiviral treatment is the most common cause of treatment failure.

Do NOT use corticosteroids as first-line agents — they may accelerate viral replication.

Quarantine the affected calf; test all herd members for viraemia; notify State and Central authorities.

Figure 2.8 — EEHV treatment protocol and monitoring/prevention framework



Left: Treatment protocol summary. Right: Monitoring, prevention and vaccine development update (mRNA EEHV vaccine currently in trial at Houston Zoo, USA).

3. Rabies in Elephants

Rabies, caused by *Lyssavirus* (negative-sense ssRNA, family *Rhabdoviridae*), is a fatal zoonotic neurological disease. While uncommon in elephants, confirmed and suspected cases have been documented in captive individuals, typically following contact with infected dogs, foxes, or bats. The invariably fatal outcome and zoonotic risk make it a priority differential diagnosis in any elephant with acute neurological signs.

3.1 Transmission and Clinical Features

- **Transmission:** Bite of a rabid dog (most common in captive elephants in India); rarely from fox or bat
- **Incubation:** Variable (weeks to months); virus travels along peripheral nerves to CNS
- **Clinical signs:** Sudden behavioural change; aggression alternating with depression; hypersalivation; dysphagia; photophobia; incoordination; seizures; coma; death
- **Differential diagnosis:** EEHV (young calves), HS, OP poisoning, musth-related aggression
- **Prognosis:** 100% fatal once clinical signs appear; no treatment available

3.2 Diagnosis

Rabies Diagnostic Protocol

- **Sample:** Brain tissue (cerebellum, hippocampus, brainstem) — ONLY sample for rabies diagnosis
- **FAT (Fluorescent Antibody Test):** Gold standard; requires fresh brain; fluorescent microscopy
- **Histopathology:** Negri bodies — eosinophilic intracytoplasmic inclusions in Purkinje cells and hippocampal neurons (H&E; ×400)
- **Seller's stain:** Rapid field stain for Negri bodies (methylene blue + basic fuchsin); Negri bodies appear magenta against blue background (×1000)
- **RT-PCR:** Most sensitive; gel electrophoresis shows diagnostic band; can use preserved tissue
- **Biosafety:** Handle brain samples with gloves, mask, and eye protection; post-exposure prophylaxis for exposed personnel

Figure 3.1 — Rabies: transmission cycle, Negri body histology and diagnostic methods

Delhi Zoo's African Elephant Died Of Viral Infection, Reveals Post-Mortem

Officials said the Encephalomyocarditis (EMCV) virus, which causes inflammation of heart muscles, was identified as the cause of death. The elephant was found dead in its enclosure on September 17.

Press Trust of India | Delhi News | Nov 04, 2025 07:00 am IST

Read Time: 2 mins

Share



Top row: Rabies transmission cycle; H&E; ×400 — Negri bodies in neurons; Seller's stain ×1000. Bottom row: FAT (Fluorescent Antibody Test) smear; RT-PCR gel showing diagnostic band.

4. Foot and Mouth Disease (FMD)

FMD is caused by *Aphthovirus* (Picornaviridae), with seven known serotypes (O, A, C, SAT1, SAT2, SAT3, Asia1). While elephants are not primary hosts, spillover from infected livestock is documented — particularly in forest fringes where elephant corridors overlap with grazing areas. FMD rarely causes death in elephants, but severe cases with secondary septicaemia can be fatal.

FMD in Elephants — Key Points

- **Clinical signs:** Vesicular lesions on feet and trunk tip; lameness; reluctance to walk; drooling if oral lesions present; fever in acute phase
- **Lesion appearance:** Fluid-filled blisters (vesicles) → rupture → raw erosions → secondary bacterial infection
- **Diagnosis:** Clinical appearance of vesicles; virus isolation; RT-PCR from vesicular fluid; ELISA for serology; probe-based antigen detection kit (IVRI Mukteswar)
- **Control:** FMD vaccination of all surrounding livestock; strict movement restrictions; lime disinfection of premises; isolation of affected animals
- **Significance in elephants:** Captive Indian elephants reported with FMD; secondary septicaemia is the main cause of mortality risk

Figure 4.1 — FMD vesicular lesions and isolation protocol



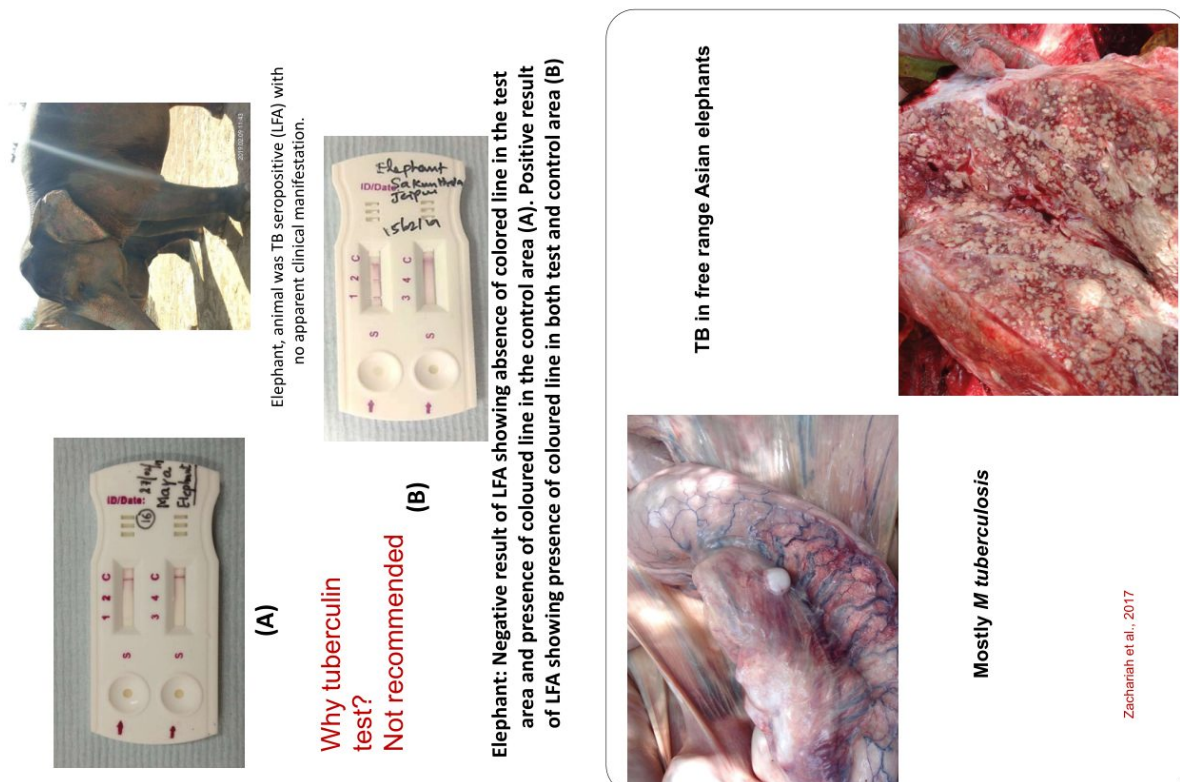
5. Poxvirus and Encephalomyocarditis Virus (EMCV)

5.1 Poxvirus (Cowpox/Elephantpox)

Elephants can be infected with cowpox virus (a member of *Orthopoxvirus*), typically through contact with infected rodents or dairy cattle. The condition is sometimes called "elephantpox" but is genetically cowpox virus. Cases are documented in zoos across Europe and in India.

- **Lesions:** Papular/vesicular eruptions on skin — trunk, ears, periocular region, feet; heal over 2–4 weeks in uncomplicated cases; gross lung lesions in severe systemic pox
- **Diagnosis:** Clinical appearance; electron microscopy of skin scrapings; PCR; histopathology (H&E; eosinophilic intracytoplasmic inclusions — Guarnieri bodies)
- **Treatment:** Symptomatic; wound care; antibiotics for secondary infection; tecovirimat (antiviral) in severe cases
- **Zoonosis:** Cowpox is a zoonotic virus; mahouts and keepers handling lesions must use PPE

Figure 5.1 — Poxvirus: vesicular skin lesions on Asian elephant



5.2 Encephalomyocarditis Virus (EMCV)

EMCV (Cardiovirus, Picornaviridae) causes acute myocarditis and sudden cardiac death, predominantly in zoo elephants. The virus is transmitted via rodent contamination of food and

water. A fatal case was documented at Delhi Zoo (India). The virus replicates in myocardial cells, causing haemorrhagic myocarditis.

- **Source:** Rats and mice (reservoir hosts); contaminated feed and water troughs
- **Clinical signs:** Sudden death or acute cardiac failure; weakness; collapse; occasionally preceded by brief respiratory distress
- **PM findings:** Haemorrhagic myocarditis; pale myocardium; pericardial effusion
- **Diagnosis:** Virus isolation; serology (ELISA); histopathology; rodent control as prevention

6. Tuberculosis (TB) in Elephants

Tuberculosis caused by *Mycobacterium tuberculosis* (human strain) is the primary mycobacterial disease in captive elephants globally. Importantly, TB in elephants is a reverse zoonosis — humans with active TB infect elephants, not vice versa in most documented cases. However, infected elephants can shed bacilli and theoretically transmit to in-contact humans (mahouts, vets). The disease is chronic, slowly progressive, and often clinically silent for years.

6.1 Diagnosis

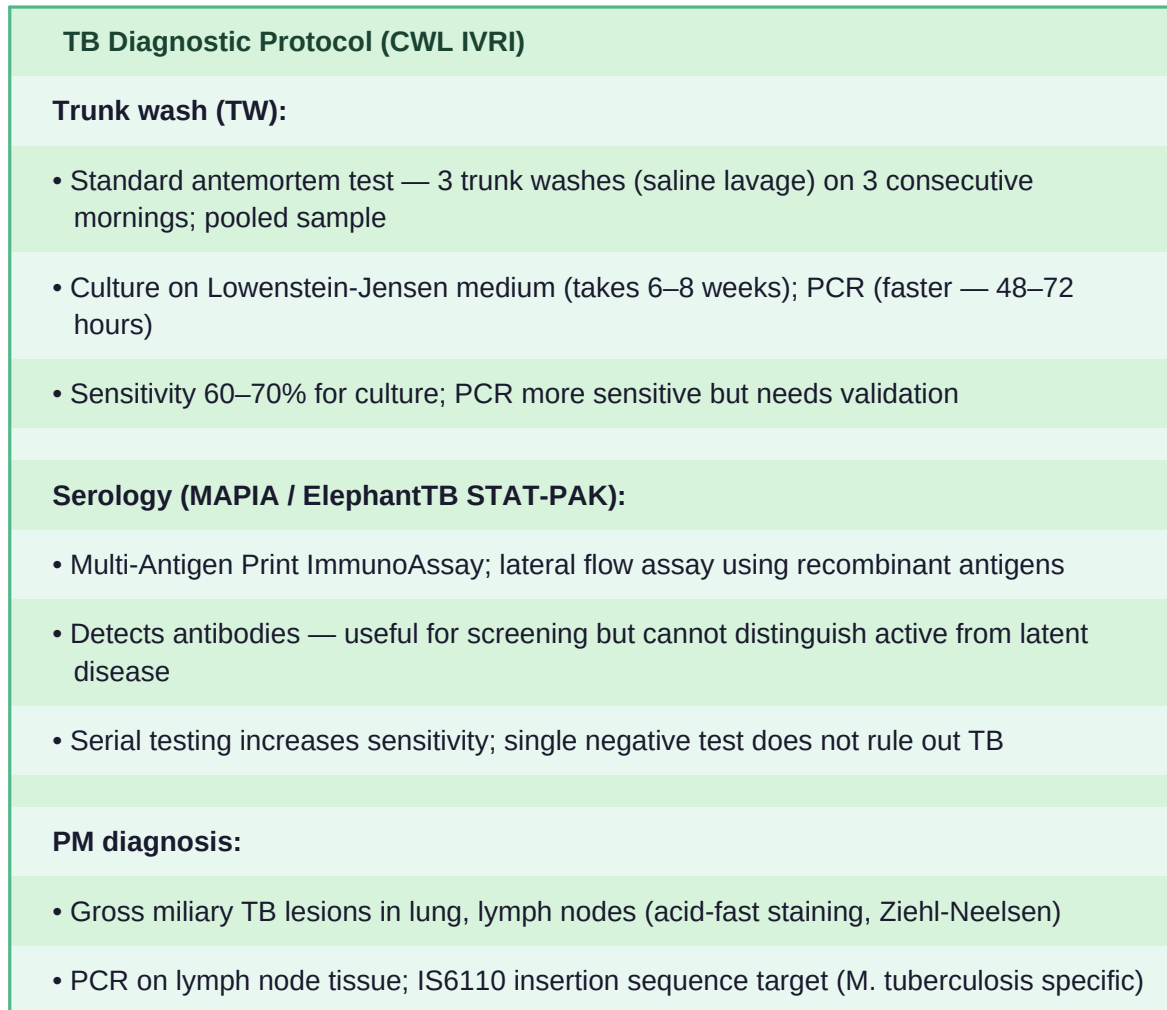
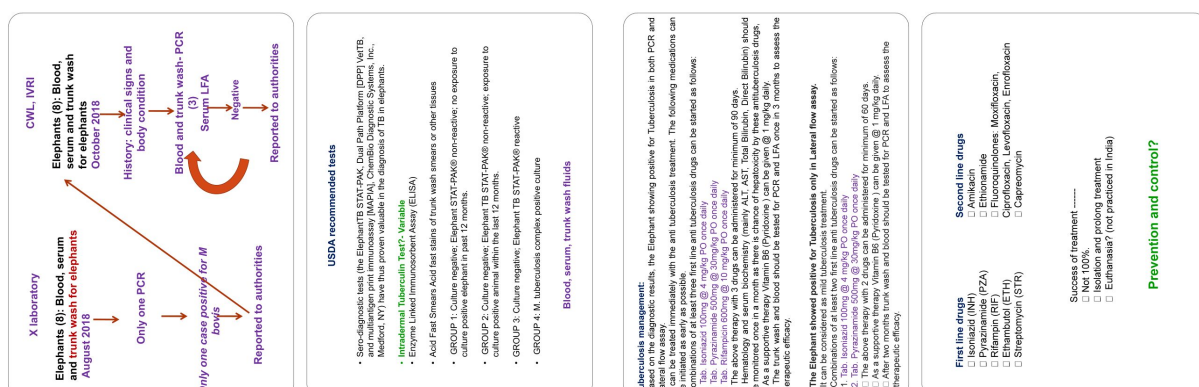


Figure 6.1 — TB diagnostics: acid-fast staining, PCR gel and LFA kit



Left: Acid-fast bacilli (blue stain) in sputum/trunk wash; IS6110 PCR gel; LFA kit (TB STAT-PAK). Right: USDA recommended diagnostic flowchart; free-range elephant lung with miliary TB lesions.

6.2 Treatment

TB Treatment Protocol (Anti-Tubercular Therapy)

- **3-drug regimen** (minimum 12 months; ideally 18–24 months for elephants):
 - Isoniazid (INH): 100 mg/day or 4–5 mg/kg
 - Pyrazinamide (PZA): 500 mg/day or 25–30 mg/kg (first 2 months)
 - Rifampicin (RIF): 600 mg/day or 10–15 mg/kg
- Drugs given in banana/jaggery boluses; observed administration (OAT) essential
- Monitor liver enzymes (AST, ALT) monthly — hepatotoxicity risk
- Pyridoxine (Vitamin B6) supplementation to prevent INH-related neuropathy
- Restrict movement and contact with other elephants during treatment

TB: Regulatory and Biosafety Requirements

TB in elephants is a notifiable disease. Report to Chief Wildlife Warden and nearest IVRI/veterinary institute.

Mahouts caring for TB-positive elephants must undergo chest X-ray and sputum test; anti-TB prophylaxis if indicated.

Three negative trunk wash cultures at monthly intervals required before declaring an elephant TB-free.

7. Anthrax

Anthrax is caused by *Bacillus anthracis*, a Gram-positive, aerobic spore-forming bacterium. It causes peracute to acute disease in elephants, with death occurring within hours of the onset of signs. Anthrax outbreaks in wildlife are typically associated with specific anthrax-prone "hot zones" where spores persist in the soil for decades.

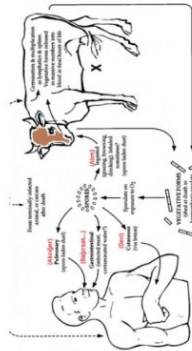
Anthrax — Key Field Points

- **Peracute/acute course:** An apparently healthy elephant found dead; or brief signs of unsteadiness, collapse, death within 1–6 hours
- **Pathognomonic gross sign:** Non-clotting, tarry (dark, watery) blood from all natural orifices (mouth, trunk, anus, vulva)
- **DO NOT open the carcass** on site — sporulation occurs on air contact; opening the carcass contaminate the entire site for 30–50 years
- **Staining:** Polychrome methylene blue (McFadyean reaction) — bamboo-rod shaped bacilli with blue capsule in blood smear; pathognomonic
- **Spore persistence:** Anthrax spores (endospores) survive in soil for 30–100 years; heavy rainfall or soil disturbance can expose buried carcasses and release spores
- **Epidemics:** Outbreaks tend to occur in hot, dry seasons followed by rainfall; documented in Joyepur Forest Range, Dibrugarh, Assam (elephants and ungulates)

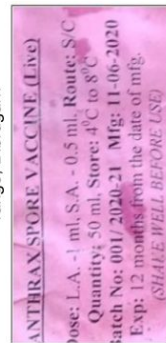
Figure 7.1 — Anthrax: carcass findings, polychrome methylene blue staining, and field investigation (Joyepur Forest Range, Assam)

Anthrax (*Bacillus anthracis*)

Three plasmid-coded exotoxins:
edema factor,
lethal toxin
protective antigen

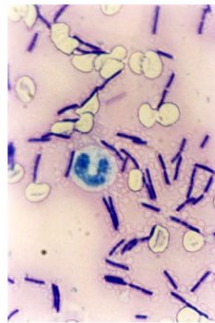


Anthrax outbreak at Joyeupur forest range, Dibrugarh



Standard Operating Procedure for Dealing with Captive and Wild Elephant Mortalities Due to Anthrax and Suspected Cases of Anthrax

- 1. Surveillance**
To improve the specific quality of and elephants, respectively, due to anthrax and the subsequent mortality, the Forest Department (FD) has initiated a surveillance system for anthrax in elephants. The FD has been instructed to report any suspected cases of anthrax to the District Collector (DC) and the Forest Officer (FO) immediately. The FD has also been instructed to take necessary steps to prevent the spread of anthrax to other elephants.
- 2. Purpose of this SOP**
The purpose of this SOP is to provide a clear and concise procedure for dealing with suspected cases of anthrax in elephants. This SOP is intended to be used by the Forest Department (FD) and the District Collector (DC) to ensure that all suspected cases of anthrax are dealt with in a timely and effective manner.
- 3. Scope of this SOP**
This SOP applies to all suspected cases of anthrax in elephants, whether they are captive or wild. It covers the entire process, from the initial reporting of a suspected case to the final disposal of the carcass. It also covers the necessary steps to be taken to prevent the spread of anthrax to other elephants.
- 4. Responsibilities**
The Forest Department (FD) is responsible for the initial reporting of a suspected case of anthrax to the District Collector (DC). The DC is responsible for the final disposal of the carcass and for ensuring that all necessary steps are taken to prevent the spread of anthrax to other elephants.
- 5. Definitions**
For the purpose of this SOP, the following definitions apply:
- **Suspected case of anthrax:** A case of anthrax in an elephant that has not been confirmed by a laboratory test.
- **Confirmed case of anthrax:** A case of anthrax in an elephant that has been confirmed by a laboratory test.



Polychrome methylene blue staining/Giemsa



ANTHRAX EMERGENCY PROTOCOL — CRITICAL

DO NOT perform a full necropsy on a suspected anthrax carcass in the field.

WHO/OIE Standard: BURN the carcass and all contaminated material; do NOT bury (burial accelerates sporulation and permanently contaminates the burial site).

All persons at scene must wear full PPE: gloves, N95 mask, goggles, boot covers.

Collect only a peripheral blood smear and blood in sealed vials for lab confirmation.

Notify Chief Wildlife Warden, IVRI, and District Collector (Section 38P, WPA 1972).

Vaccinate all livestock in a 5-km radius with Sterne strain anthrax vaccine immediately.

8. Haemorrhagic Septicaemia (HS)

Haemorrhagic Septicaemia is caused by *Pasteurella multocida* type B:2 (and occasionally E:2). HS is primarily a disease of bovids (cattle, buffalo), but elephants are susceptible — particularly captive/semi-captive animals under stress. The disease is peracute and characterised by sudden high fever, respiratory distress, and death within 24–48 hours. Outbreaks in elephants have been investigated in Kalahandi district, Odisha and Kartapat WLS.

HS — Diagnosis and Management

- **Predisposing factors:** Physical stress (translocation, overwork, musth), nutritional deficiency, heavy rainfall season, contact with infected cattle
- **Clinical signs:** High fever (40–42°C); oedematous swelling of head/neck/brisket; respiratory distress; profuse salivation; tympanitis; death in 24–48 hours
- **PM findings:** Severe haemorrhagic pneumonia; peracute congestion of all organs; haemorrhagic lymph nodes; subcutaneous gelatinous oedema
- **Histopathology:** Bipolar organism (oval shape with bipolar staining — "safety pin" appearance) on Gram stain or Leishman stain
- **Best diagnostic sample:** Bone marrow aspirate from femur or rib (taken within hours of death before lysis; highest yield)
- **Treatment:** Penicillin G 20,000 IU/kg IM; Tetracycline; Sulphonamides; fluid therapy; anti-inflammatory (Meloxicam)
- **Prevention:** HS vaccination of all livestock in a 2-km radius; separate elephant water sources from cattle wallows

Figure 8.1 — HS: bipolar organisms (Leishman stain), haemorrhagic lung lesions, and Kalahandi field investigation



Left: Bipolar staining of *P. multocida*; gross PM lung lesions; haemorrhagic lymphadenopathy. Right: Field investigation at Kartapat WLS — trackers, necropsy team, and carcass disposal operations.

Figure 8.2 — HS control: livestock vaccination and bone marrow sampling protocol



Control measures

1. Vaccination against HS for domestic animals
2. Carcass disposal
3. Available Samples
4. Awareness

Heart blood, LN, Lung
Bone marrow is the ideal sample



Molecular characterisation and serotyping of *Pasteurella multocida* isolates from Asiatic elephants (*Elephas maximus*)

2016

Chandranak, Busavegowdanadoddi Marnaik | Shivashankar, Beechugondahalli Papanna | Girdhar, Papanna | Nagaraju, Doddegowdanakoppalu
Maniesowda

We report the cultural and molecular characterization of *Pasteurella multocida* isolates recovered from Asiatic elephants at Bandipur tiger reserve forests in Karnataka state, India. The first herd comprised consecutive draughts from 2010 to 2017, and sudden heavy monsoon rains in 2012, twenty-two elephants died during this period of heavy rain and between July and July 2017, 100% of the elephants died. Autopsies revealed the presence of lesions in the lungs and liver of some dead elephants and lesions of soft tissue in femur bones were seen in several carcasses, while fresh samples were not available. Bone marrow samples of femur bones were aseptically collected. Heart blood and spleen samples collected at postmortem and two out of seven bone marrow samples yielded the growth of *P. multocida*. The isolates were typed as *P. multocida* type 8. The phylogenetic analysis of the isolates was carried out by 16S ribosomal rRNA (rRNA) gene based PCR and sequencing of 1.3 kbp nucleotides on the 16S rRNA gene of the isolates. The isolates showed highest sequence identity with *P. multocida* isolates of bovine and caprine origin. This is the first report of molecular study on *P. multocida* from Asiatic elephants.

Bone marrow and Lymph nodes

9. Clostridial Infections

Clostridial diseases are caused by Gram-positive, anaerobic, spore-forming bacilli of the genus *Clostridium*. Several species are pathogenic in elephants, producing potent toxins that cause rapid multiorgan failure:

Species	Disease	Key Features
<i>C. difficile</i>	Pseudomembranous enterocolitis	Watery/haemorrhagic diarrhoea; pseudomembrane on colonic mucosa; associated with antibiotic use
<i>C. perfringens</i> (Type A/D)	Enterotoxaemia	Sudden death; bloat; haemorrhagic enteritis; toxin detection by ELISA from intestinal contents
<i>C. tetani</i>	Tetanus	Muscular rigidity; lockjaw; opisthotonus; triggered by wound infection; rare in elephants
<i>C. botulinum</i>	Botulism	Flaccid paralysis; toxin from contaminated feed; peracute death possible
<i>C. chauvoei</i>	Blackleg (rare)	Gas gangrene of muscle; crepitation on palpation; acute febrile death

Field Note: Clostridial Sample Collection

When handling intestines of suspected Clostridial cases: do NOT open intestinal loops in the field. Intestinal contents must be sealed and sent to laboratory immediately under cold chain.

Clostridial toxins dissipate rapidly after death — intestinal samples must be collected within 1–2 hours of death for toxin ELISA to be meaningful.

Figure 9.1 — Clostridial: *C. difficile* enterocolitis (gross mucosa), tetanus, and published case literature

Clostridial infections

Enterotoxaemia, Tetanus, *Clostridium difficile* Enterocolitis



***Clostridium perfringens* type A beta2 toxin in elephant (*Elephas maximus indicus*) and pygmy hog (*Sus salvanius*) with haemorrhagic enteritis in Assam, India**

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Samples need to be collected and transported in anaerobic conditions

CLOSTRIDIUM INFECTIONS IN ASIAN ELEPHANTS (*Elephas maximus*)

Clostridium spp. are anaerobic, spore-forming bacteria that produce potent toxins. Spores persist in the environment and can cause severe disease in elephants.

1. <i>Clostridium novyi</i>	2. <i>Clostridioides difficile</i> (formerly <i>Clostridium difficile</i>)	3. <i>Clostridium perfringens</i>	4. <i>Clostridium tetani</i>
Cause: Severe tissue infection and toxemia 	Cause: Enterocolitis (inflammation of the intestines) 	Cause: Enterotoxigenic, necrotic enteritis 	Cause: Tetanus
How it causes disease: - Spores enter through wounds or damaged environments. - Germinate in anaerobic (low oxygen) environments. - Produce toxins (alpha toxin, beta toxin) causing tissue destruction, edema, shock, and may lead to sudden death. Clinical signs: - Swelling, pain, and necrosis. - Fever, depression, and discharge. - Sudden death in acute cases.	How it causes disease: - Overgrowth occurs when normal gut flora is disrupted (e.g., after antibiotic use). - Produce toxins A and B. - Toxin A causes inflammation, leading to colitis and diarrhoea. - Toxin B causes severe tissue damage and leads to fluid loss, dehydration, and shock. Clinical signs: - Watery or mucoid diarrhea. - Straining, abdominal discomfort. - Depression, weakness.	How it causes disease: - Common inhabitant of the large intestine. - Overgrowth and sudden toxin production. - Toxins (alpha, beta, epsilon, iota, etc.) cause necrosis, gas production, and can cause sudden death. Clinical signs: - Acute diarrhea. - Colic, abdominal pain. - Toxicosis, depression, rapid breathing. - Sudden death in severe cases.	How it causes disease: - Spores enter through wounds. - Germinate in anaerobic conditions. - Produce tetanospasmin toxin, causing muscle rigidity and spasms. Clinical signs: - Muscle stiffness, rigidity. - Tetanus (lockjaw), difficulty swallowing. - Severe spasms; recumbency may be fatal.
TRANSMISSION & RISK FACTORS - Spores are resistant to heat, cold, and the environment. - Spores are found in soil, manure, and feces. - Risk increases with stress, poor hygiene, and antibiotic treatment, or poor diet.	DIAGNOSIS - Clinical examination. - Fecal examination (ELISA). - Post-mortem examination (histopathology, PCR). - Neutropeny findings.	TREATMENT - Antibiotics (penicillin, ampicillin). - Supportive care: fluids, electrolytes, NSAIDs. - Fluid and electrolyte management.	PREVENTION - Good hygiene and sanitation. - Proper wound care. - Judicious use of antibiotics. - Prophylactic use of antibiotics. - Tetanus toxoid vaccination for all elephants.

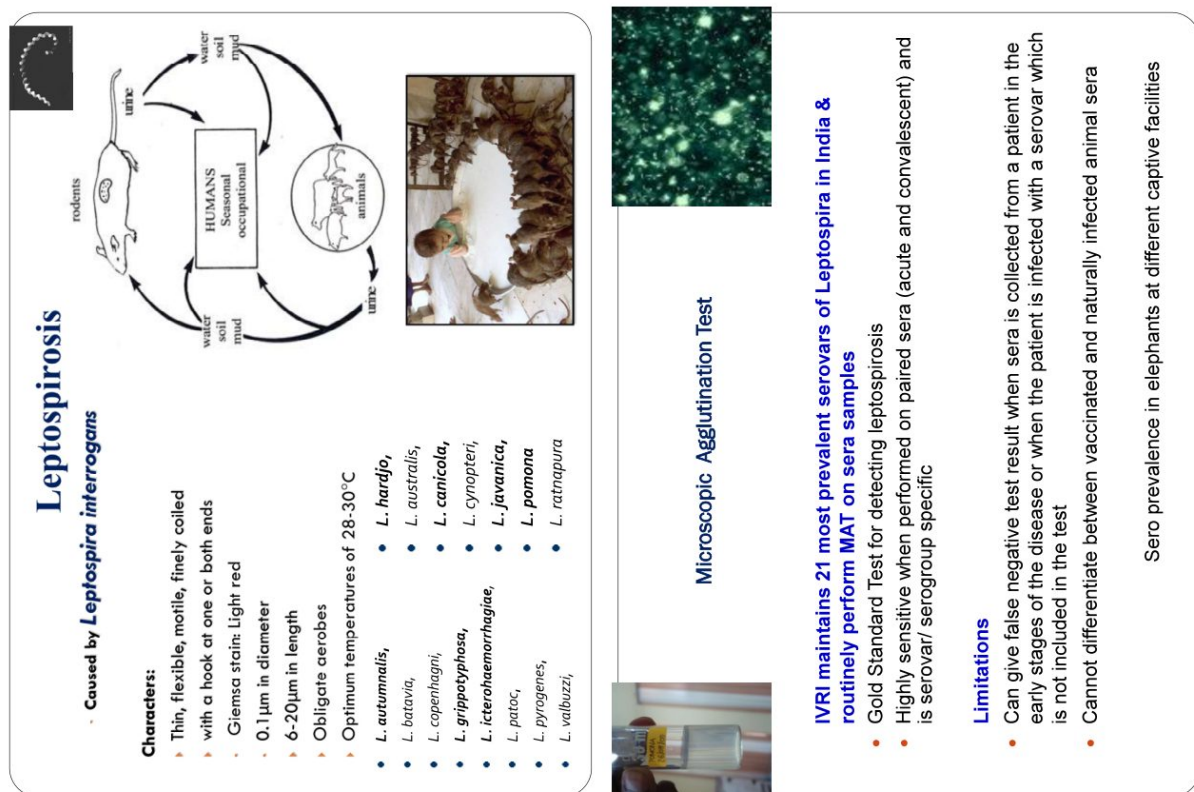
Early detection and prompt treatment are critical, as clostridial disease can progress rapidly and be fatal in elephants.

10. Leptospirosis

Leptospira spp. (spirochete bacteria) cause leptospirosis, a globally important zoonosis. The disease is transmitted through water or soil contaminated by urine of infected animals (rodents, cattle, dogs). Elephants are exposed at contaminated water bodies, river banks, and during monsoon flooding. The IVRI Leptospirosis Laboratory maintains 21 serovars for diagnosis.

- **Transmission:** Contact with contaminated water/soil through skin abrasions, mucous membranes, or ingestion; rodents are primary reservoir; cattle/dogs amplify hosts
- **Clinical signs in elephants:** Fever; icterus (jaundice); haemoglobinuria (red urine); haemorrhagic manifestations; abortion in females; in severe cases, acute renal failure and death
- **Diagnosis:** MAT (Microscopic Agglutination Test) — paired samples 2 weeks apart; 4-fold rise in titre diagnostic; Latex Agglutination Test (LAT) for field screening; darkfield microscopy of urine; PCR on blood/urine
- **Treatment:** Penicillin G (effective if early); Doxycycline 5 mg/kg; supportive fluid therapy
- **Prevention:** Avoid water body access during monsoon floods; rodent control; vaccination of working elephants (available in some countries)

Figure 10.1 — Leptospirosis: transmission cycle, MAT serovars panel, and Latex Agglutination Test



11. Parasitic Diseases

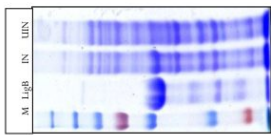
Parasitic diseases in elephants encompass a wide range of helminths, protozoa, and ectoparasites. While rarely causing acute mortality in healthy adults, parasitic burdens contribute to debilitation, secondary infections, and increased susceptibility to other diseases — particularly in captive elephants on restricted diets or in poor body condition.

11.1 Helminth Parasites

Parasite	Type	Location	Clinical Significance
<i>Fasciola jacksoni</i>	Trematode (Liver fluke)	Bile ducts / Liver	Fasciolosis: hepatomegaly, biliary fibrosis, emaciation; <i>Fasciola</i> eggs with bile in faeces
<i>Pseudodiscus collinsi</i>	Amphistome (rumen fluke)	Large intestine / Caecum	Heavy burdens → colitis; diarrhoea; seen on PM as reddish-brown flukes attached to mucosa
<i>Strongylus</i> spp. / <i>Strongyles</i>	Nematode	Anterior/posterior intestine	Colitis, protein loss; larvicidal migration causes hepatic tracking
<i>Toxocara</i> (in calves)	Nematode	Small intestine	Visceral larva migrans; pot-belly; poor growth in calves
<i>Oxyuris</i> (pinworm)	Nematode	Rectum	Tail rubbing; anal irritation; common in captive elephants

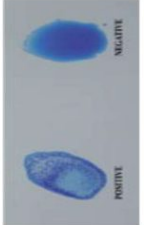
Figure 11.1 — Helminth parasites: *Pseudodiscus collinsi* (amphistome), GI nematodes (*Strongyle*), and *Fasciola jacksoni* eggs and flukes

Latex Agglutination Test



Purified LigB, Lipl32 and Lipl41 proteins are routinely employed in Latex Agglutination Test to screen field sera samples for detecting Leptospirosis and the results are compared with MAT

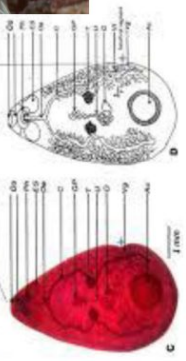
- LAT employs only OMP of *Leptospira* not the whole pathogen
- Serves as a pen site diagnostic test
- Able to detect infection caused by any pathogenic leptospiral serovars



Pseudodiscus collinsi is a digenetic trematode (amphistome fluke)







11.2 Protozoal Parasites

Protozoal Parasites

- **Balantidium coli/suis:** Large ciliated protozoan; infects colon; causes yellow diarrhoea; photomicrograph shows cysts and trophozoites; Treatment: Metronidazole 15–25 mg/kg BID for 5 days + Omeprazole
- **Haemoparasites:** Trypanosoma spp. found in blood smears; transmitted by biting flies; associated with emaciation, anaemia, lymphadenopathy; treated with Diminazene aceturate
- **Sarcocystis:** Intramuscular cysts; incidental finding at necropsy in most cases
- **Amoeba:** Entamoeba spp. occasionally detected in faecal smears; significance unclear in elephants

Figure 11.2 — Protozoal parasites: *Balantidium* cyst microscopy, yellow diarrhoea case, blood smear with haemoparasite



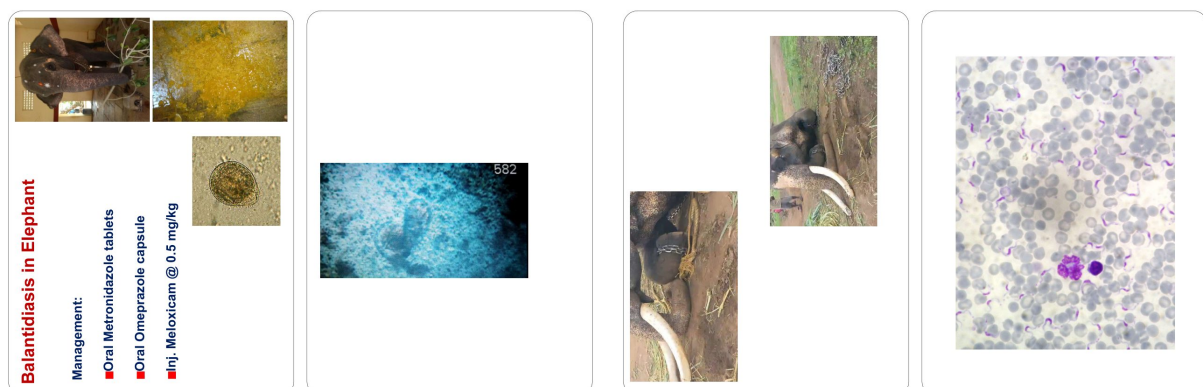
12. Surgical and Non-Infectious Conditions

Beyond infectious disease, field and zoo veterinarians frequently encounter traumatic, degenerative and management-related conditions. These require hands-on surgical skills and a good knowledge of elephant anatomy:

Common Surgical and Management Conditions

- **Sinus wound / abscess:** Common in captive elephants; secondary to bullhook injuries, trauma or tusk injuries; requires debridement, irrigation with povidone-iodine, and systemic antibiotics; may need surgical debridement under chemical restraint (Etorphine / Medetomidine)
- **Corneal opacity / eye conditions:** Corneal ulcers from foreign bodies, trauma or infection; treatment: topical antibiotic + atropine; assess retinal integrity; ophthalmoscopy under chemical restraint
- **Foot problems:** Extremely common in captive elephants — overgrown nails, foot pad cracks, sole abscesses; caused by hard substrate, poor hygiene, inadequate exercise; treatment: trimming, foot soaks (copper sulphate/betadine), systemic antibiotics
- **Dystocia:** Rare but life-threatening; calf mal-presentation; requires controlled delivery under sedation; pre-positioning of delivery assistance equipment recommended at breeding herds
- **Colic / Impaction:** Abdominal pain; anorexia; reduced faecal output; treat with liquid paraffin per os, fluids, Buscopan; severe cases may require flank examination under sedation

Figure 12.1 — Surgical cases: sinus wound treatment (Dudhwa NP), corneal opacity, foot nail overgrowth and pad cracks



Left: Sinus wound irrigation and debridement in elephant Roopakali (Dudhwa NP). Right: Corneal opacity assessment; overgrown foot nails; sole cracks in captive elephant.

13. Emerging Threats and Mass Mortality Events

Wildlife disease ecology is constantly evolving. RNA viruses, climate-driven pathogen range shifts, and novel toxicological threats pose growing risks to elephant populations. Veterinarians and forest officers must remain alert to unusual mortality clusters that may signal emerging pathogen activity.

Emerging Threats to Asian Elephant Health

- **Cyanobacteria / Algal bloom toxins:** Botswana 2020 — largest documented elephant mass mortality event; 350 elephants died around waterholes; cyanobacterial toxins (suspected nodularin/microcystin or BMAA) implicated; elephants showed neurological signs before collapse
- **Monkeypox (now Mpox):** Documented in captive elephants from Africa; can spillover to keepers; vesicular skin lesions; zoonotic potential elevated since smallpox vaccination stopped
- **Novel RNA viruses:** Bat-origin coronaviruses (SARS-CoV-2 antibodies detected in some captive elephants in Belgium); Paramyxoviruses; Nipah (bats → elephants theoretically possible in Kerala)
- **Climate change:** Shifts in mosquito vectors → altered arbovirus range; changes in grass/crop phenology affecting mycotoxin risk; increased heat stress mortality events
- **Antibiotic resistance:** Elephants in close proximity to human settlements accumulate resistant bacteria; antibiotic sensitivity testing mandatory before treatment

Figure 13.1 — Botswana 2020: mass elephant mortality around waterhole (cyanobacteria)

Successful treatment of sinus wound in Asiatic elephant at Dhudhwa National Park

Elephant Roopakali

ABST - Amikacin, Streptomycin and colistin



Corneal opacity and cataract



Cracks in the foot pad

Over grown nails with cracks

14. Field Practice Principles and Preparedness

Effective disease diagnosis and management in the field requires structured protocols, appropriate equipment, and a systematic approach. The following principles should guide field veterinarians at any elephant casualty or sick elephant call:

14.1 Antibiotic Use and Sensitivity Testing

- Always culture-based treatment when possible — empirical antibiotics only in emergencies
- Collect swabs/tissue/blood for culture BEFORE starting antibiotics
- Disk diffusion / Kirby-Bauer method for sensitivity testing (readily available at district labs)
- Avoid broad-spectrum antibiotic overuse — accelerates resistance in zoonotic bacteria
- Document all drug use: drug name, dose, route, duration, outcome — essential for case analysis

14.2 Documentation and Photography Protocol

- **Photograph before touching:** Establish GPS location; photograph the scene and carcass position before examination
- **Video documentation:** Record gross PM findings continuously; voice narrate observations
- **Systematic PM photography:** Each body cavity; each organ; all lesions with scale bar
- **Labelling:** Sample labels in waterproof ink; photograph each labelled sample with ID visible
- **Chain of custody:** Log each sample handed over; signatures required
- **Necropsy report:** Submit within 48 hours using standard DFO/CWL format

14.3 Essential Field Kit

Category	Items
Personal Protective Equipment	Nitrile gloves (×3 pairs/person), N95 mask, splash goggles, Tyvek suit, boot covers, biohazard waste bags
Sample Collection	EDTA tubes, plain red-top tubes, formalin pots (10%), sterile swabs, viral transport media, bone marrow aspiration needle, labelled zip-lock bags
Instruments	Scalpel (large handle + #22 blades), bone saw or hacksaw, forceps, scissors, ruler, weighing scale
Diagnostics	Portable glucometer, haemoglobin meter, glass slides, Wright/Giemsa stain, Seller's stain, darkfield microscopy (if available)

Communication	GPS unit, camera with macro lens, waterproof field notebook, contact list (IVRI, lab, WW office, PCCF)
Cold Chain	Ice packs, insulated cool box, cold chain labels

15. Conclusion and Key Take-Aways

The field of elephant health medicine in India is rapidly evolving, driven by increasing human–elephant interface, expanding captive elephant populations, and the emergence of novel pathogens. This chapter has presented the major infectious, parasitic, and management-related diseases of Asian elephants, with emphasis on field-relevant diagnostic and treatment protocols.

Key Take-Aways for Field Veterinarians

1. **EEHV** is the highest-priority disease for calves and young elephants — famciclovir must be pre-positioned; early clinical recognition saves lives.
2. **Anthrax** requires zero-necropsy protocol in the field — burn all material; never bury.
3. **Tuberculosis** is a chronic silent killer and a two-way zoonotic risk — mandatory surveillance of working elephants and their handlers.
4. **Rabies** — brain is the only diagnostic sample; full PPE mandatory; post-exposure prophylaxis for exposed personnel.
5. **HS and Clostridial** — bone marrow is the gold-standard postmortem sample; collect before lysis.
6. **Parasites** — regular faecal examination and strategic deworming are the cornerstone of captive elephant health maintenance.
7. **Documentation** — systematic photography, GPS recording and rapid lab submission are as important as the clinical intervention itself.
8. **Preparedness** — every elephant camp and zoo should maintain a preparedness kit, an emergency contact tree, and a supply of pre-positioned drugs (famciclovir, anthrax vaccine, HS vaccine, broad-spectrum antibiotics).

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Figure 15.1 — Acknowledgements page from the source presentation



Boswana-Cyanobacteria (HCN)

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Thank you

